

Introduction

- In clinical trials, the ICH E9(R1) estimands framework requires sponsors to pre-specify a strategy for every intercurrent event (ICE).
- For continuous endpoints, the composite strategy integrates ICE occurrence directly into the endpoint: subjects who experience an ICE receive a fixed failure value v in place of their unobserved outcome.
- Unlike binary composites, however, **no regulatory guidance exists for selecting v on a continuous scale.**

In practice, failure values are often fixed arbitrarily (e.g., $v = 0$ mL FEV₁). Bell et al. demonstrate that such extreme values induce severe bimodal variance inflation, increasing required sample sizes by up to 4-fold. They propose **control distribution failure definitions (CDFD)**: anchoring v to a quantile of the concurrent control completer distribution (e.g., the 10th percentile), producing a population-level, clinically interpretable benchmark.

- CDFD resolves the arbitrariness problem but introduces a new challenge: because v depends on unknown control distribution parameters, it carries **estimation uncertainty that must be propagated through to inference on δ** (the target estimand).

- Under the Bayesian framework, uncertainty in the control distribution parameters propagates to the posterior for δ through the joint likelihood automatically, without requiring separate variance corrections such as the delta method or bootstrap.

We develop two such Bayesian estimators for composite strategy treatment effects under CDFD.

Background

Consider a two-arm RCT with continuous endpoint y and binary ICE indicator z . Under the composite strategy, each subject receives a composite outcome:

$$y_i^* = \begin{cases} y_i & z_i = 0 \text{ (completer)} \\ v & z_i = 1 \text{ (ICE occurred)} \end{cases}$$

Rather than fixing v arbitrarily, we define it as the **q -th quantile of the control completer distribution**, the control distribution failure definition (CDFD):

$$v = \nu + \Phi^{-1}(q) \omega,$$

where ν and ω are the mean and standard deviation of outcomes among control completers.

- For example, $q = 0.10$ treats ICE occurrence as equivalent to performing at the 10th percentile of the standard of care.

The target estimand is the marginal composite treatment effect

$$\delta = \mathbb{E}_1[y^*] - \mathbb{E}_0[y^*].$$

Because ν and ω are unknown, v carries estimation uncertainty that must propagate through to inference on δ .

Methodology

Parametric Joint Model

- We jointly model the continuous outcome and ICE occurrence for all N subjects simultaneously, estimating control distribution parameters and therefore v concurrently with the treatment effect.
- Posterior uncertainty in v propagates automatically to the final posterior for δ .

Outcome model (completers):

$$y_i \sim N(\beta_0 + \beta_1 tr_i + \beta_2 b_i, \sigma^2)$$

ICE model (all subjects):

$$\text{logit}(p_i) = \gamma_0 + \gamma_1 tr_i + \gamma_2 b_i$$

Any pre-specified baseline predictors of dropout can be incorporated into the ICE submodel.

Weakly informative priors reflecting the FEV1 scale are assigned to all parameters. The joint posterior $\pi(\theta \mid D)$ — where θ denotes the full parameter vector and D the observed trial data — is obtained via MCMC.

Treatment Effect

- Because δ is a nonlinear function of parameters from models with distinct link functions, the marginal treatment effect cannot be read directly from regression coefficients.
- We recover it via G-computation, a standardization procedure that recovers marginal treatment effects by marginalizing counterfactual model predictions over a target covariate distribution.

At each posterior draw $\theta^{(s)}$ we standardize over the observed covariate distribution:

$$\mu_j^*(\theta) = \frac{1}{N} \sum_{i=1}^N [(1 - p_{ij})\mu_{ij} + p_{ij}v(\theta)],$$

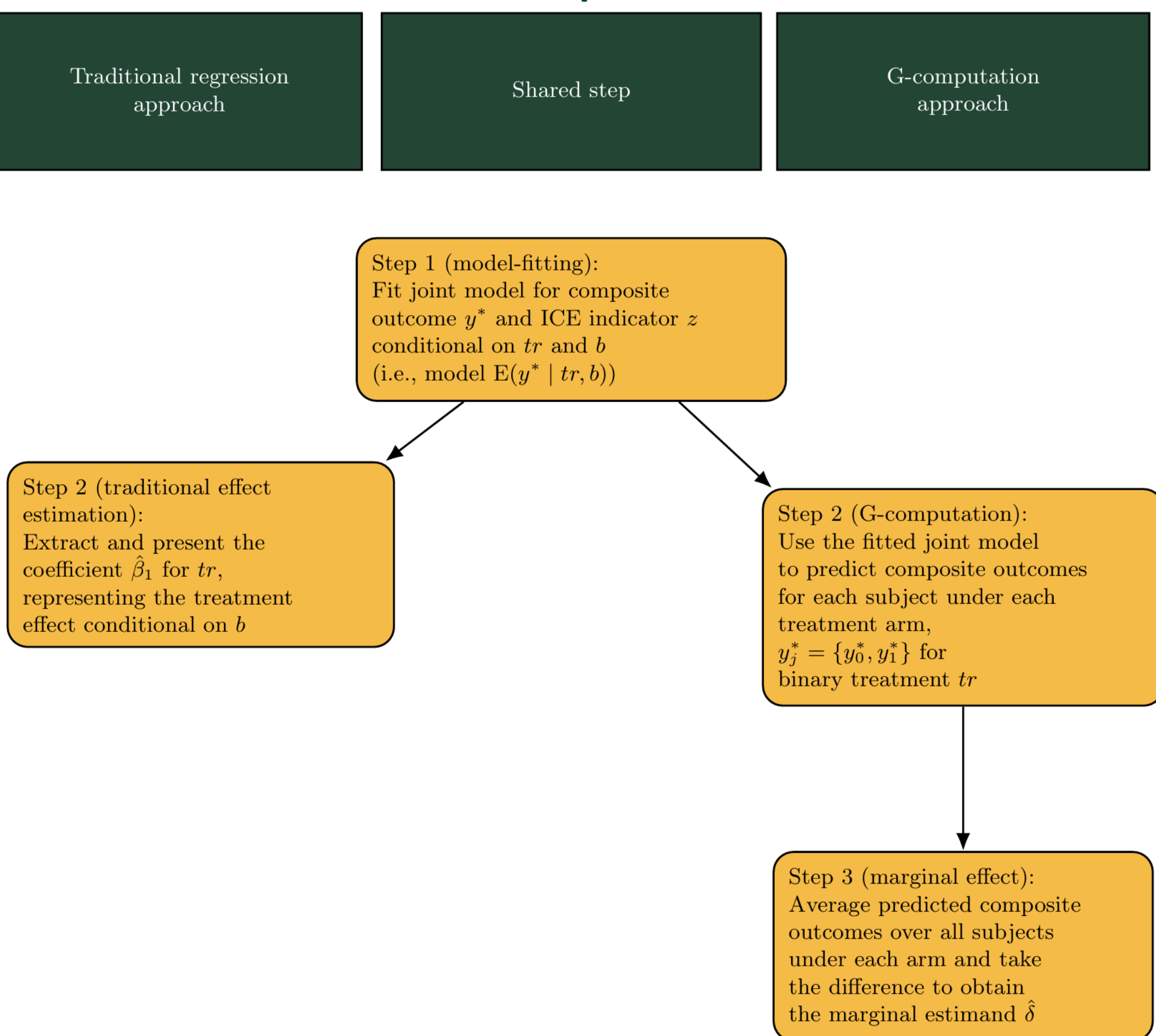
where μ_{ij} and p_{ij} are the model-predicted outcome mean and ICE probability for subject i under treatment j , evaluated counterfactually.

The failure value at each draw is computed from the marginal control completer parameters:

$$v(\theta) = \nu(\theta) + \Phi^{-1}(q) \omega(\theta).$$

Since $\delta(\theta)$ is a deterministic function of θ , evaluating it at each MCMC draw $\theta^{(s)}$ yields a corresponding set of draws that approximates the marginal posterior $\pi(\delta \mid D)$.

G-Computation



Competing Methods

Method	Approach
Plug-in	Treats v as fixed; ignores failure value uncertainty entirely
Delta method	First-order Taylor expansion to propagate uncertainty in v analytically
Bootstrap	Re-estimates v from resampled control arm within each resample
Frequentist MI	Draws M plausible v values from sampling distribution; pools via Rubin's Rules
Two-Stage Bayes	Stage 1 estimates a full posterior for v from control completer outcomes; Stage 2 propagates this uncertainty through Bayesian multiple imputation

Results

We evaluate all methods via simulation based on a parallel two-arm asthma trial with FEV₁ (liters) as the continuous endpoint. Data are generated under a normal linear ANCOVA model with $\beta_0 = 0.3$, $\beta_2 = 1.0$, $\sigma = 0.20$, and baseline FEV₁ $\sim N(2.0, 0.4^2)$. The failure value uses $q = 0.10$, giving $v = 1.727$ L under the true population parameters. Results are based on 5000 replications per scenario.

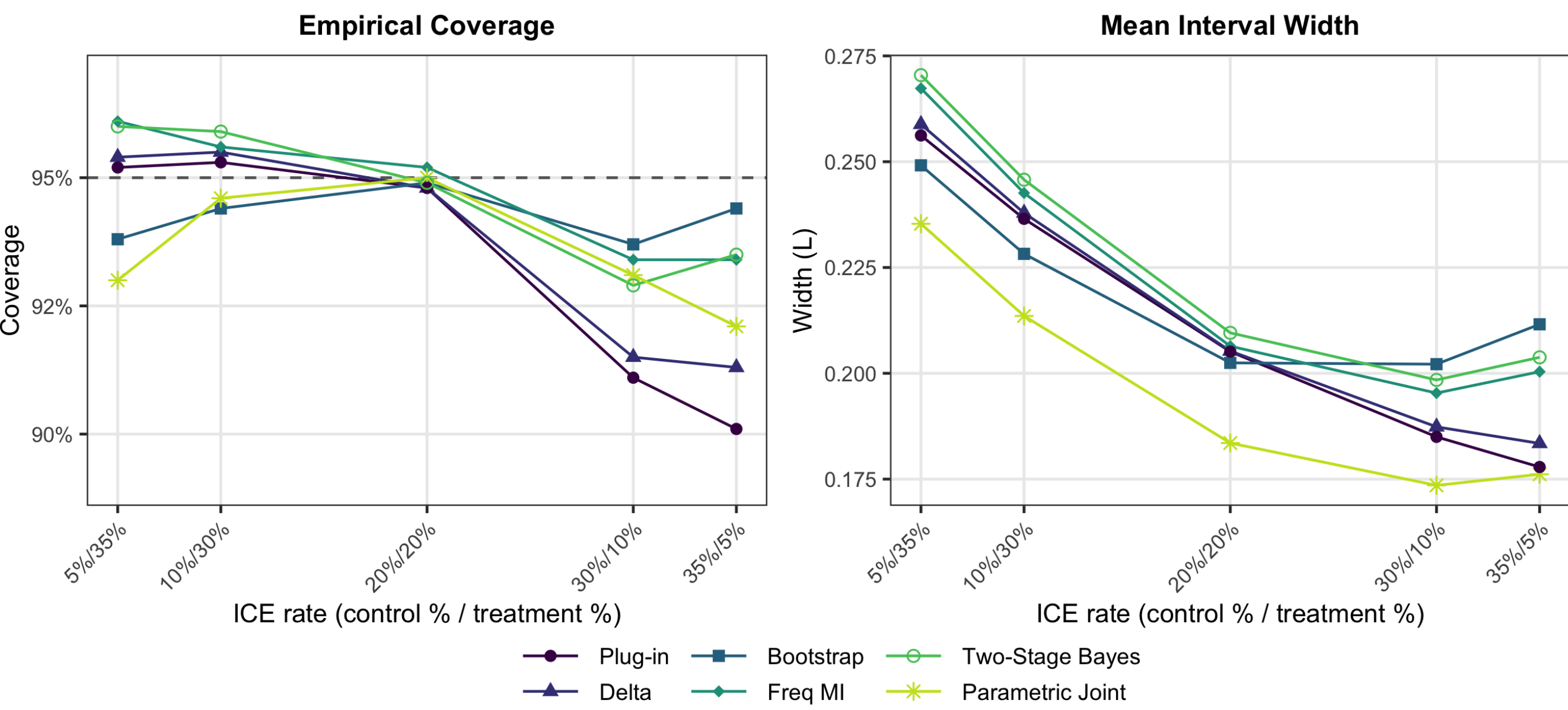


Figure 2: Coverage and interval width by asymmetric ICE rate pair (control%/treatment%), $n = 100$ per arm.

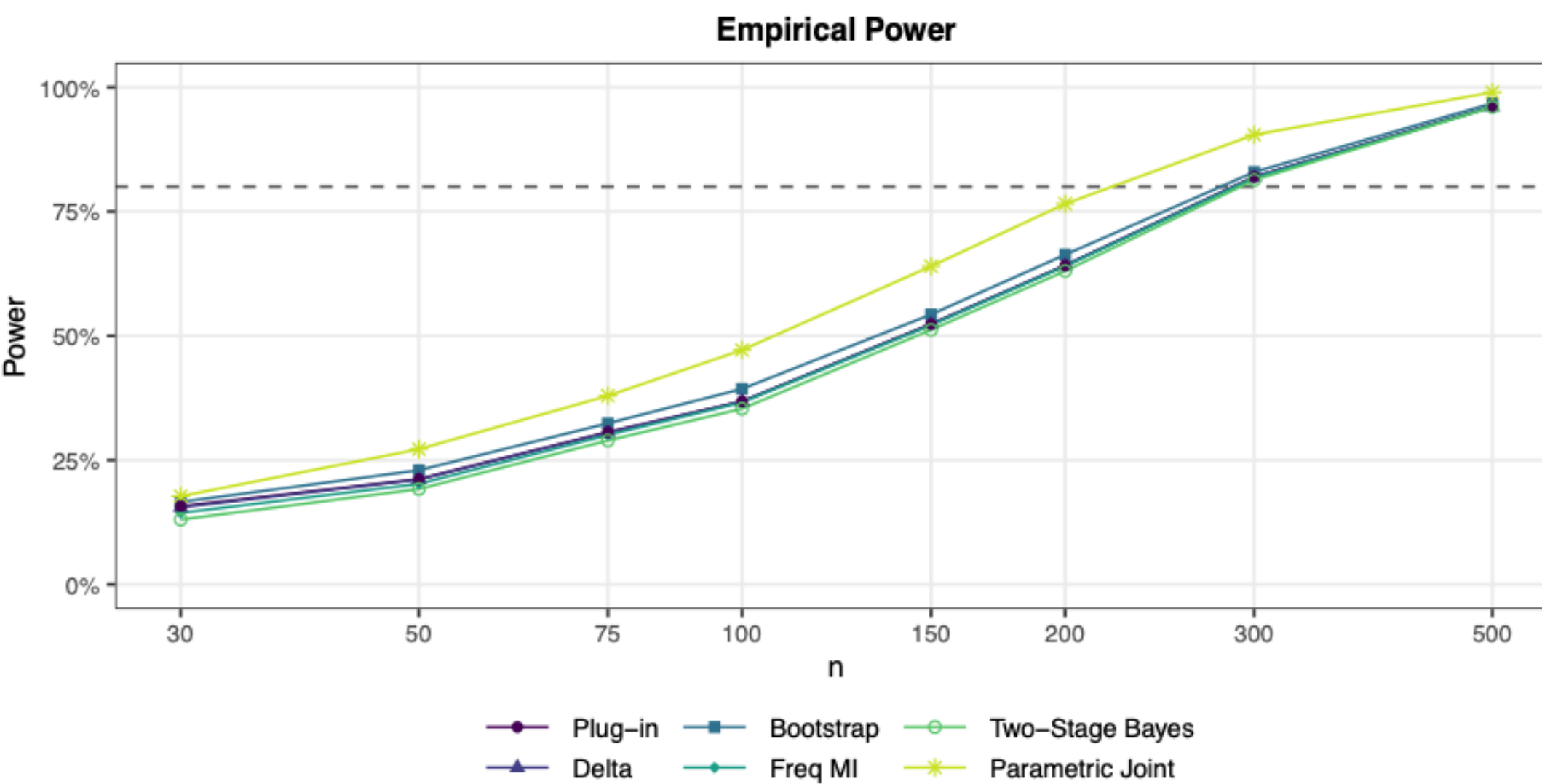


Figure 3: Empirical power as a function of sample size per arm (n) at symmetric ICE rates, MAR assumption ($\delta = 0.09$ L, 30%/30% ICE rates, $q = 0.10$, $K = 5,000$ replicates).

Discussion

Our two Bayesian estimators anchor the failure value to the control distribution (CDFD) and carry its estimation uncertainty automatically through to δ without requiring separate variance corrections.

The parametric joint model consistently achieves narrower credible intervals than all competing methods, leveraging ICE probability information from the full study population rather than treating ICE occurrence as a deterministic binary switch. The two-stage Bayesian estimator maintains near-nominal coverage across all scenarios and offers a natural mechanism for incorporating historical control data through informative priors on ν and ω .

- The parametric joint model is preferred when dropout is driven by observable covariates and efficiency is a priority
- The two-stage estimator is the more conservative default for confirmatory trials where the dropout mechanism is uncertain